

LJUBLJANA BRNIK, Slovenia

WMO: **140140**

Lat: **46.218N** Lon: **14.473E** Elev: **1273** StdP: **14.03** Time Zone: **1.00 (EUC)** Period: **94-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	1	9.3	14.1	4.6	7.3	16.1	8.7	9.0	20.4	16.1	36.9	13.2	33.9	1.9	290	0.473	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	7	22.9	87.8	69.6	84.7	68.5	82.1	67.6	71.4	83.7	69.9	81.7	68.3	79.5	6.2	120	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	67.5	105.9	77.3	65.9	100.0	75.5	64.4	95.0	73.5	36.0	84.0	34.6	82.0	33.4	79.8	77.9	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	13.4	11.2	9.3	DB	2.8	92.0	6.6	3.4	-2.0	94.5	-5.8	96.5	-9.5	98.4	-14.3	100.9	(4)
(5)				WB	2.2	74.3	6.4	1.9	-2.4	75.7	-6.2	76.8	-9.8	77.9	-14.5	79.3	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours		DBAvg	49.8	30.7	33.7	41.3	49.6	57.8	64.9	68.0	67.4	58.8	50.8	41.6	32.0	(6)
(7)			DBStd	14.73	7.53	7.26	6.75	5.96	5.53	5.81	5.13	5.25	5.61	6.63	7.76	7.76	(7)
(8)			HDD50	2312	598	457	275	78	6	0	0	0	3	73	264	558	(8)
(9)			HDD65	5865	1063	877	735	461	231	72	29	35	196	442	701	1023	(9)
(10)			CDD50	2242	0	0	5	66	247	448	559	540	268	96	13	0	(10)
(11)			CDD65	321	0	0	0	0	7	70	123	109	12	0	0	0	(11)
(12)			CDH74	3615	0	0	0	14	179	819	1333	1140	128	2	0	0	(12)
(13)			CDH80	1051	0	0	0	0	24	223	425	368	11	0	0	0	(13)
(14)	Wind		WSAvg	3.1	2.6	2.9	3.6	3.7	3.7	3.6	3.4	3.1	2.8	2.7	2.6	2.3	(14)
(15)	Precipitation		PrecAvg	59.6	3.2	3.5	4.0	4.4	4.6	5.6	5.2	5.3	6.2	6.5	6.7	4.6	(15)
(16)			PrecMax	83.5	10.1	12.0	9.2	8.0	9.0	9.1	10.0	10.2	15.4	17.6	17.5	11.1	(16)
(17)			PrecMin	46.4	0.2	0.2	0.5	0.3	1.5	1.9	1.9	1.1	1.6	0.2	0.4	0.0	(17)
(18)			PrecStd	8.7	2.1	2.8	2.4	1.9	1.8	1.6	2.0	2.5	3.2	3.8	3.7	2.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.9	59.7	67.8	76.1	82.8	89.3	91.8	92.9	81.7	72.4	62.5	53.7		(19)
(20)			MCWB	47.1	46.6	51.9	58.1	65.1	69.7	70.2	70.5	67.5	62.1	55.7	50.2		(20)
(21)		2%	DB	48.1	52.3	62.8	71.4	78.7	85.8	87.8	87.7	77.8	68.3	57.9	49.7		(21)
(22)			MCWB	44.2	44.2	50.1	55.4	62.8	68.7	69.8	69.6	65.6	60.4	53.9	46.8		(22)
(23)		5%	DB	44.9	48.5	58.6	67.8	75.3	82.4	84.6	84.1	74.0	64.8	55.5	46.2		(23)
(24)			MCWB	42.1	42.9	48.3	54.3	61.2	67.3	68.5	68.7	63.6	58.2	52.5	44.4		(24)
(25)		10%	DB	42.2	44.9	54.2	64.0	71.7	79.0	81.7	80.5	70.5	61.6	53.0	42.7		(25)
(26)			MCWB	40.1	40.6	46.0	52.8	59.5	65.7	67.8	67.5	61.7	56.5	50.5	41.4		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.4	48.5	53.8	60.2	67.1	72.6	73.5	73.5	69.0	63.4	57.0	51.1		(27)
(28)			MCDB	50.1	55.1	63.6	72.2	79.0	84.5	87.2	86.4	78.7	70.3	60.1	53.2		(28)
(29)		2%	WB	45.1	45.8	51.2	57.3	64.3	70.2	71.5	71.1	66.6	61.4	54.8	47.4		(29)
(30)			MCDB	47.2	50.6	60.4	68.0	76.0	82.3	84.1	83.2	75.8	67.2	57.5	49.0		(30)
(31)		5%	WB	42.8	43.7	49.3	55.2	62.2	68.3	69.9	69.6	64.8	59.4	53.0	44.3		(31)
(32)			MCDB	44.2	47.9	56.4	64.9	72.8	79.9	81.6	81.3	72.5	63.7	55.4	45.6		(32)
(33)		10%	WB	39.9	41.0	47.5	53.4	60.4	66.6	68.3	68.0	62.8	57.4	50.8	41.3		(33)
(34)			MCDB	41.4	44.4	53.2	62.1	69.6	77.4	79.4	79.0	69.1	61.2	52.8	42.2		(34)
(35)	Mean Daily Temperature Range		MDBR	13.4	17.2	20.9	21.6	22.3	21.8	22.9	22.3	19.8	17.9	12.8	11.9		(35)
(36)		5% DB	MCDBR	17.7	24.3	30.1	30.8	29.5	27.7	28.2	28.6	25.0	22.4	17.0	16.3		(36)
(37)			MCWBR	13.3	16.5	18.4	16.7	15.0	13.0	12.4	12.5	13.9	14.3	12.4	13.1		(37)
(38)		5% WB	MCDBR	15.0	21.9	25.3	26.6	26.7	26.1	26.0	26.2	22.6	19.6	14.2	14.0		(38)
(39)			MCWBR	12.0	15.5	16.1	14.9	14.0	12.8	12.3	12.4	12.9	13.0	10.6	11.7		(39)
(40)	Clear-Sky Solar Irradiance		taub		0.318	0.346	0.393	0.443	0.444	0.452	0.449	0.443	0.419	0.393	0.353	0.317	(40)
(41)			taud		2.402	2.325	2.225	2.114	2.160	2.186	2.203	2.232	2.266	2.308	2.374	2.450	(41)
(42)			Ebn at Noon		246	259	261	258	262	260	259	255	250	238	228	233	(42)
(43)			Edh at Noon		23	30	38	47	46	45	44	41	36	30	23	20	(43)
(44)	All-Sky Solar Radiation		RadAvg		402	659	1028	1349	1654	1865	1876	1644	1160	754	416	335	(44)
(45)			RadStd		59	106	141	160	163	125	125	179	151	87	64	50	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.99	N/A	N/A	+1.91	+0.97	+0.99	N/A	-288	+189	+119	(46)
(47)	Regional (77 neighbors)	+0.86	N/A	N/A	+1.44	+0.81	+0.63	-144	-229	+171	+112	(47)

Nomenclature: See separate page